WASP-39b

R-1622

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-39_160320` · created 2026-08-02T04:44:24+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457467.867 +/- 0.04 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.056
Transit depth (Rp/Rs)^2	3.54 +/- 2.09 [%]
Orbital Inclination (inc)	86.95 +/- 2.11
Airmass coefficient 1 (a1)	1.25 +/- 0.78
Airmass coefficient 2 (a2)	-0.2 +/- 0.61
Scatter in the residuals of the lightcurve fit is	59.3 %
Best Comparison Star	#2 - [46, 382]
Optimal Aperture	2.41
Optimal Annulus	5.25
Transit Duration (day)	0.092 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.188 ± 0.056 vs 0.1457 (deviation 0.76σ)
Duration fit vs published	0.092 d vs 0.12 d (deviation 0.74σ)
Mid-time O–C	-100.0 min
Depth SNR	3.4
Residual scatter	59.3 %

Light curve

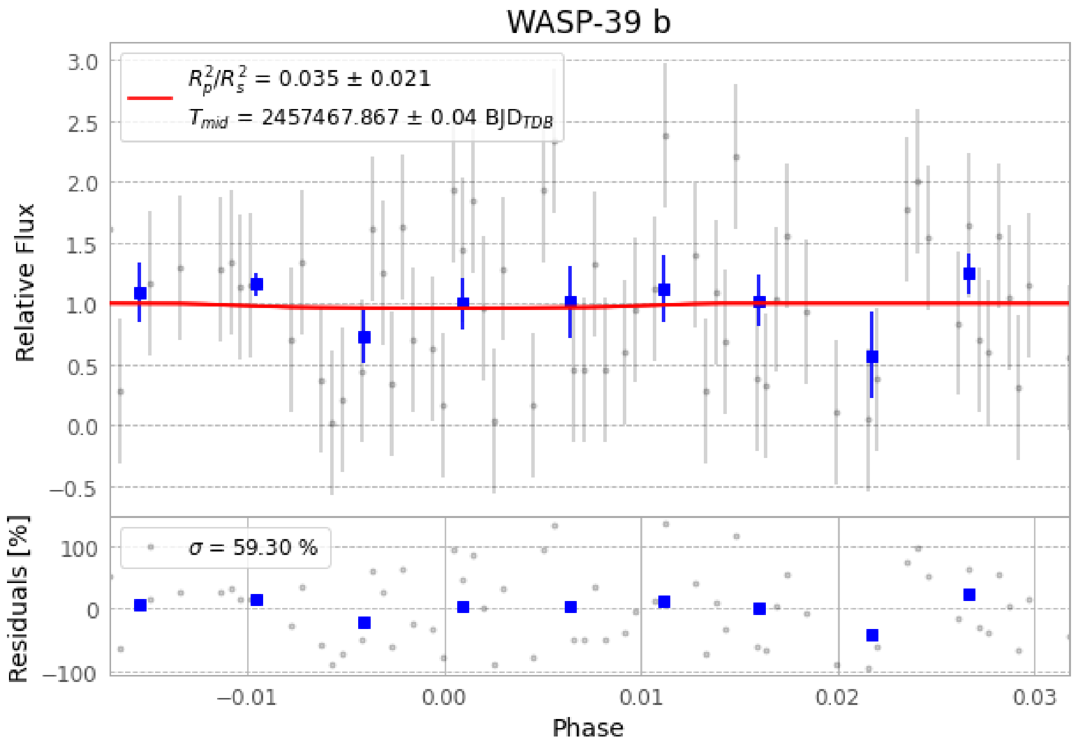

Figure 1 — FinalLightCurve_WASP-39 b_2016-03-19.png (SHA-256 847eb744fd82a23e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 214], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 51dc8f13e94d1b9b…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

99 FITS frames (65 MB), `WASP-39160320065712.FITS` → `WASP-39160320115112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-160320.log (cb8235105c75…), FinalLightCurve_WASP-39 b_2016-03-19.png (847eb744fd82…), FinalLightCurve_WASP-39 b_2016-03-19.csv (cc41c27e9425…)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)